

# Industrial Automation Systems

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## Computing and Mathematics

May, 2026

# Industrial Automation Systems (A14182)

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|---------------------|--------------------------------|---------------|--------------|
| <b>Short Title:</b> | Industrial Automation Systems  |               |              |
| <b>Faculty</b>      | Science and Computing          |               |              |
| <b>Department</b>   | Computing and Mathematics      |               |              |
| <b>Cluster</b>      | Automotive, Automation and IoT |               |              |
| <b>Author</b>       | JDALY                          |               |              |
| <b>Credits</b>      | 5                              | <b>Level:</b> | Intermediate |

## Description of Module / Aims

Industrial Automation Systems are pervasive in the Manufacturing and Service Industries. Systems vary from small turn-key systems to large integrated systems complete with Supervisory Control and Data acquisition/analysis. In this module, students will use industry-standard industrial controllers, networking and User-Interface technologies required to develop Automated Systems. Safety considerations will also be covered.

## Programmes

|           |                                              |           |
|-----------|----------------------------------------------|-----------|
| COMP-0665 | BSc (Hons) in Applied Computing (WD_KACCM_B) | 2 / 4 / E |
| COMP-0665 | BSc (Hons) in Applied Computing (WD_KCOMP_B) | 2 / 4 / E |
| COMP-0665 | BSc (Hons) in Computer Science (WD_KCMSC_B)  | 2 / 4 / E |

## Indicative Content

- Industrial Automation Systems: Overview; architectures; standards; development methodologies
- Process specification and automation design
- Programming, Commissioning and Testing of solutions to Automated Control problems
- Integrating hardware and software based safety systems
- Automation of systems using Analog Sensors and Transducers; Typically, involving closed-loop control
- Human-Machine Interfacing, Supervisory Control and Data Acquisition and Networking

## Learning Outcomes

*On successful completion of this module, a student will be able to:*

1. Appraise the characteristics, options and application of Industrial Automation Systems in batch and continuous process systems.
2. Analyse the hardware/interface requirements when controlling different categories of industrial process.
3. Develop embedded control software on Industry standard Industrial Control Systems.
4. Design, commission and test event-driven solutions to common Industrial Automation Processes.
5. Analyse the safety requirements of different Industrial Automation scenarios.
6. Use industry standard Human-Machine Interface and networking technologies.

## Learning and Teaching Methods

- Combination of lectures and lab-based practicals.
- The lectures will cover the theory and underlying technologies behind Industrial Automation Systems.
- The lab-based practicals, building on the theoretical knowledge from lectures, provide the practical skills to design, model, simulate and implement Industrial Automation Systems.

- The practical content will use industry standard technologies and tools to design, model and implement Industrial Automation Systems.
- Student will be encouraged to enhance their lab work and assessment submissions using self-directed research and learning into Industrial Automation Systems.

## Assessment Methods

|                            | Weighing | Outcomes Assessed |
|----------------------------|----------|-------------------|
| Continuous Assessment      | 100%     | 1,2,3,4,5,6       |
| <i>Lab Report</i>          | 30%      | 2,3,4,5,6         |
| <i>In-Class Assessment</i> | 20%      | 1,4,5             |
| <i>Assignment</i>          | 50%      | 2,3,4,6           |

## Assessment Criteria

**<40%:** Inability to demonstrate simple software and hardware designs from user requirements.

**40%-49%:** Able to specify interfacing and software requirements for simple discrete control systems.

**50%-59%:** In addition, be able to implement industrial controller based solutions for simple discrete control systems.

**60%-69%:** In addition, be able to develop optimised designs for a specific industrial control system.

**70%-100%:** In addition, combine self-directed research of Industrial Automation Systems in assessment work. Exhibit the ability to solve unforeseen problems through the use and modification of self-learned skills and tools.

## Learning Modes

| Learning Mode        | F/T Hours | P/T Hours |
|----------------------|-----------|-----------|
| Lecture              | 24        |           |
| Practical            | 24        |           |
| Independent Learning | 87        |           |

## Requested Resources

- COMPUTER LAB: BYOD Lab